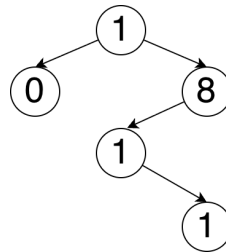


Task Description | End-Sem Practical Examination | EE-B | Batch 1

Write a C program that accepts the **last 5 digits** of your own 11-digit roll number as an array of digits. For example, if the roll number is **25295918011**, then, the last five digits are **18011** and it should be stored as {1, 8, 0, 1, 1}. The program should perform Binary Search Tree operations, Heap construction, Heap deletion, and Heap traversal/printing based on the roll number.

Tasks to be performed

1. Input the last 5 digits of the roll number into an integer array (take user input).
2. Preserve the original copy of the array for later use.
3. Insert the digits in the original order into a BST using the following rule.
 - a. If the new digit is smaller than the current node, insert it in the left subtree.
 - b. If the new digit is greater than or equal to the current node, insert it in the right subtree.



4. Print the following traversals of the BST:
 - a. Inorder Traversal (For instance, **01118** is the inorder output for the given example)
 - b. Post-Order Traversal (For instance, **01181** is the post-order output for the given example)
5. Using the preserved original array, create a Heap based on the following condition.
 - a. if `roll_number % 2 == 0`, create a Min Heap
 - b. if `roll_number % 2 == 1`, create a Max Heap
6. Print the heap array after heap construction (Min Heap: **01181** / Max Heap: **81011**).
7. Delete the root node of the heap once and heapify again.
8. Print the deleted element.
9. Print the new heap array after deletion and heapification.

Note: Equal elements are allowed as parent and child in Min Heap / Max Heap in this problem.

Important: Each print output must be on a new line. If the given example outputs do not contain whitespaces between digits, then your output must also not contain whitespaces.